Skin Disease Classification

Applied Deep Learning — Group Project Report

Group 3 — Summer Semester 2026

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1 Project Setting and Goals

1.1 Problem Context

[To be written by Thilina]

1.2 Project Objectives

[To be written by Thilina]

1.3 Related Work

[To be written by Thilina]

2 Dataset Description and Exploratory Analysis

2.1 Dataset Overview

The dataset is a publicly available collection of dermatological images sourced from Kaggle, originally comprising 22 classes of skin conditions with a total of 15,444 images split into predefined training (13,898) and test (1,546) sets. Each class corresponds to a clinically recognised skin condition, ranging from common disorders such as acne and eczema to less prevalent conditions such as candidiasis and vasculitis. All images are provided as RGB photographs with heterogeneous resolutions, as the dataset aggregates clinical and dermoscopic images from multiple sources.

The predefined train/test split follows an approximate 90/10 ratio across all classes, indicating a stratified partitioning strategy. No official validation split is provided, and no patient-level identifiers are available, which limits the ability to enforce patient-level separation between splits. The dataset does not include metadata such as patient demographics, anatomical location or imaging device, which restricts the scope of bias analysis.

2.2 Class Reduction: From 22 to 18 Classes

An initial transfer learning experiment using all 22 classes with ResNet18 and the identical training protocol described in Section 3.3 yielded a balanced accuracy of 0.652 and a macro F1 of 0.637. Analysis of per-class recall revealed systematic weaknesses that motivated a principled reduction to 18 classes. Four classes were removed based on complementary clinical and statistical criteria:

Unknown/Normal, with 1,651 training images and a recall of 0.921, functions as a catch-all category for images that do not depict a specific skin disease. While the model classifies it with high recall, its inclusion introduces label noise. Visual inspection confirmed that this category contains non-dermatological images such as household objects and animals, indicating systematic label noise beyond clinical ambiguity. Furthermore, "normal" skin appearance varies widely across skin tones, ages and body regions, and the category boundary is defined by exclusion rather than by shared visual features. Removing it eliminates a source of distributional bias that disproportionately inflates overall accuracy without contributing to the diagnostic task.

Sun/Sunlight Damage, with 312 training images and a recall of 0.588, represents the early stage of a clinical continuum that includes actinic keratosis, a class retained in the dataset. The visual overlap between chronic sun damage and early actinic keratoses is substantial, as both present as rough, scaly patches on sun-exposed skin. Retaining both classes forces the model to learn a distinction that even dermatologists resolve through biopsy rather than visual inspection alone. With only 312 training images, the class lacks sufficient coverage to learn this subtle boundary.

Moles, with 361 training images and a recall of 0.525, present a classification challenge because the distinction between benign melanocytic naevi and early melanoma is the subject of dedicated binary classification systems in the literature. In a multi-class setting with 22 heterogeneous conditions, the model cannot learn the fine-grained features required for this distinction. At 0.525 recall, nearly half of mole images are misclassified, predominantly into the Skin Cancer and Benign Tumors categories. Including this class risks creating a false sense of diagnostic capability for a task that requires specialised architecture and data.

Lupus, with 311 training images and a recall of 0.471, encompasses multiple clinical subtypes: discoid lupus erythematosus, subacute cutaneous lupus and the malar rash of systemic lupus. These subtypes present with markedly different morphology. With only 311 images, the dataset cannot adequately represent this heterogeneity, resulting in the lowest recall among all classes. The visual overlap with rosacea, as both present with facial erythema, further compounds the classification difficulty.

After removal, the reduced dataset comprises 18 classes with 11,263 training images and 1,249 test images. The class reduction improved balanced accuracy from 0.652 to 0.699, a gain of 4.7 percentage points, and macro F1 from 0.637 to 0.695, a gain of 5.8 points, under identical training conditions. This confirms that the removed classes were net contributors to classification noise rather than to diagnostic signal.

2.3 Class Distribution and Imbalance

Table 1 presents the distribution of images across the 18 retained classes. The largest class, Benign Tumors with 1,093 training images, contains approximately 4.4 times as many samples as the smallest, Candidiasis with 248 images. This moderate imbalance ratio is characteristic of medical imaging datasets, where disease prevalence varies across conditions.

Table 1: Class distribution in the 18-class dataset, sorted by training set size.

Class	**Train**	**Test**	**Train %**
Benign Tumors	1,093	121	9.7
Eczema	1,010	112	9.0
Tinea	923	102	8.2
Psoriasis	820	88	7.3
Actinic Keratosis	748	83	6.6
Vitiligo	714	82	6.3
Skin Cancer	693	77	6.2
Acne	593	65	5.3
Warts	580	64	5.1
Lichen	553	61	4.9
Drug Eruption	547	61	4.9
Vascular Tumors	543	60	4.8
Infestations/Bites	524	60	4.7
Bullous	504	55	4.5
Vasculitis	461	52	4.1
Seborrheic Keratoses	455	51	4.0
Rosacea	254	28	2.3
Candidiasis	248	27	2.2
Total	**11,263**	**1,249**	**100.0**

To address class imbalance during training, we applied inverse-frequency weighting to the cross-entropy loss function: each class receives a weight proportional to the reciprocal of its sample count, normalised so that the weights sum to the number of classes. This approach

penalises misclassification of minority classes more heavily without altering the training data distribution.

Figure 1 visualises this distribution.

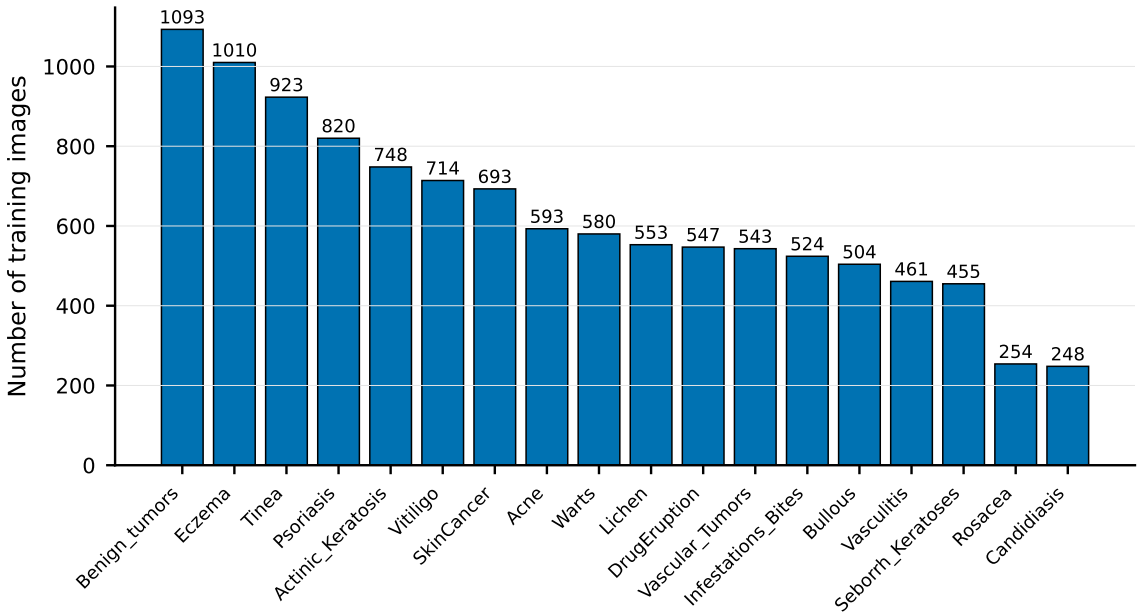

Figure 1: Class distribution across the 18 retained classes, sorted by training set size.

2.4 Image Properties

All images are stored in RGB format with heterogeneous resolutions ranging from approximately 30 to over 4,000 pixels in both width and height, with a median of approximately 600 pixels. The majority of images have aspect ratios close to 1:1, which permits uniform resizing to 224×224 pixels for transfer learning models and 32×32 for traditional ML baselines with only moderate distortion. Per-class analysis of mean channel intensities shows that classes such as Vitiligo and Rosacea exhibit characteristic colour signatures, while classes such as Tinea and Eczema overlap substantially in colour space.

2.5 Visual Inspection

Figure 2 displays representative training images for each class. Several observations are relevant for classification:

- High intra-class variability: conditions such as Infestations/Bites and Drug Eruption encompass visually disparate presentations, ranging from insect bites to scabies and from maculopapular to urticarial reactions.
- Inter-class similarity: Eczema and Psoriasis both present as erythematous, scaly patches; Actinic Keratosis and Skin Cancer share morphological features on sun-damaged skin; Bullous diseases and Drug Eruptions can both manifest as blistering lesions.
- Imaging heterogeneity: the dataset mixes clinical photographs, including full-body and regional views, with close-up dermoscopic images, introducing scale and context variability within classes.
- Distinctive classes: Vitiligo with its characteristic depigmentation, Acne with comedones and pustules on facial skin, and Warts with verrucous texture exhibit strong visual signatures that facilitate classification.

Figure 2: Sample images from the training set (first four images per class). The visual heterogeneity within and between classes is evident.

3 Methodology and Challenges

3.1 Preprocessing

[To be written by Diana]

3.2 Baseline Models

[To be written by Diana]

3.3 Transfer Learning

[To be written by Diana]

3.4 Evaluation Metrics

[To be written by Diana]

3.5 Challenges

[To be written by Diana]

4 Results, Conclusion, and Outlook

4.1 Model Comparison

[To be written by Nahidul]

Table 2: Model comparison on the test set. Best values per metric in bold.

Metric	**RF Baseline**	**ResNet18**	**EfficientNet-B0**
Accuracy	0.364	**0.694 (0.669–0.720)**	0.681 (0.656–0.708)
Balanced Acc.	0.284	**0.699 (0.673–0.725)**	0.694 (0.668–0.721)
F1 (macro)	0.308	**0.695 (0.668–0.720)**	0.685 (0.658–0.710)

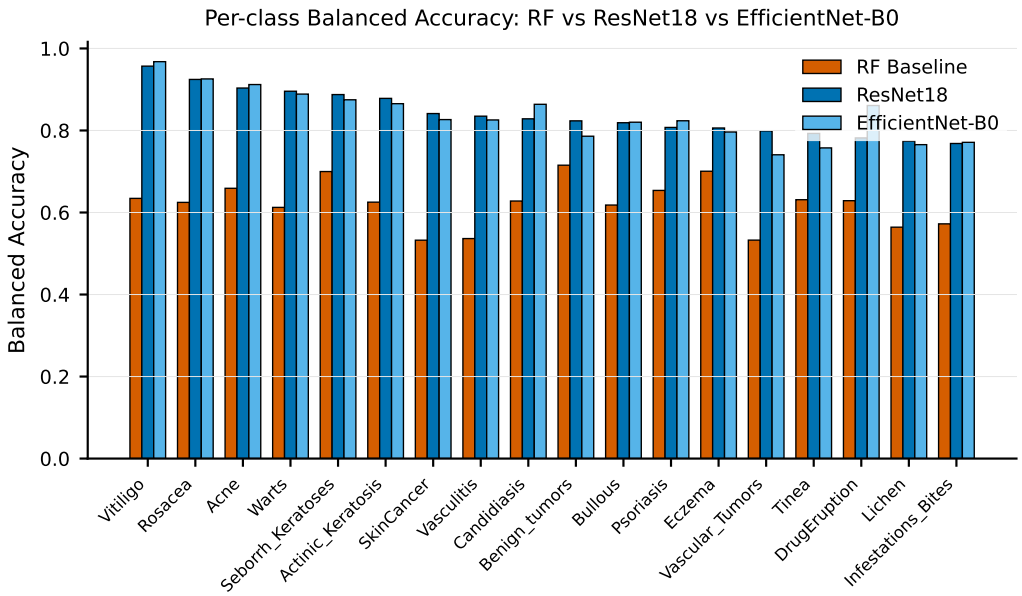

Figure 3: Per-class balanced accuracy (one-vs-rest) across all three models on the 18 retained classes.

4.2 Per-Class Analysis

[To be written by Nahidul]

4.3 Confusion Matrix Analysis

[To be written by Nahidul]

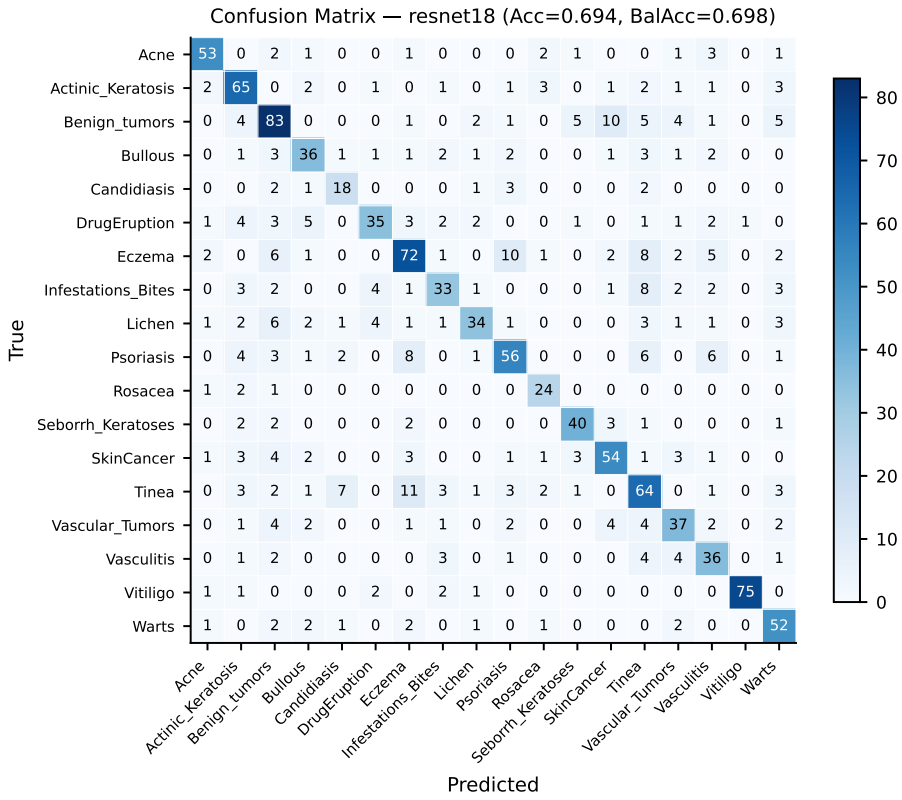

Figure 4: Confusion matrix for ResNet18 on the 18-class test set.

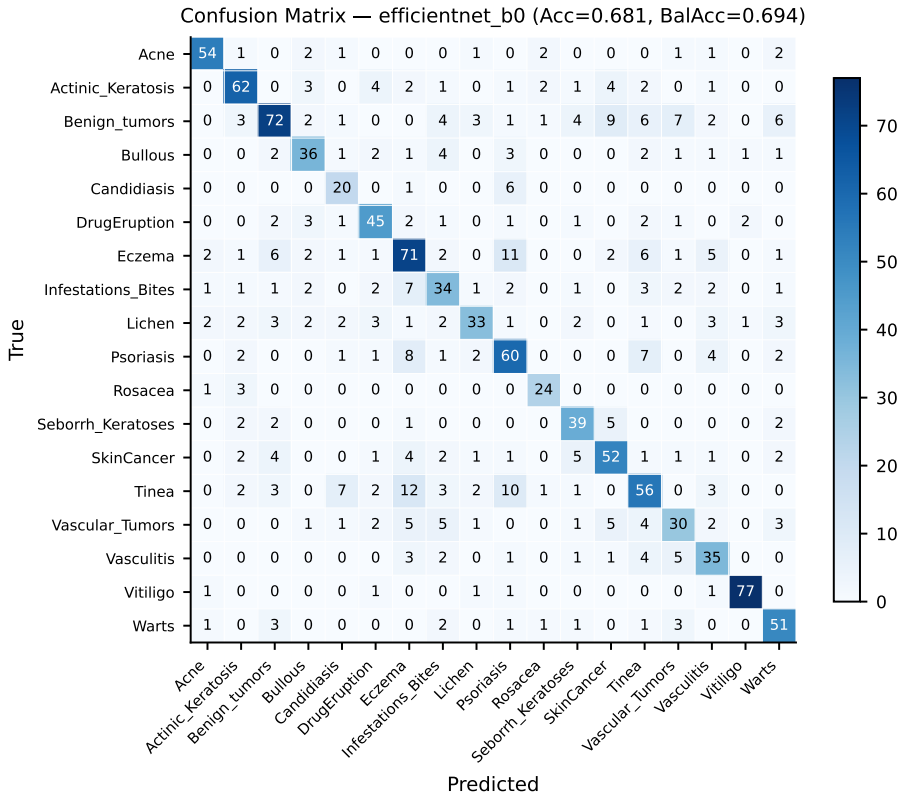

Figure 5: Confusion matrix for EfficientNet-B0 on the 18-class test set.

4.4 Conclusion

[To be written by Nahidul]

4.5 Ethical Considerations

[To be written by Nahidul]

4.6 Outlook

[To be written by Nahidul]

A AI Tool Disclosure

This project used Claude Code (Anthropic) as a coding assistant for script development, exploratory data analysis and report structuring. All architectural decisions, clinical reasoning for class reduction and interpretation of results were made by the team members. Generated code was reviewed, tested and modified before inclusion in the pipeline.

B Work Distribution

Member	Contribution
Thilina Ranmal Angunna Gamage	Project motivation, related work, literature review
David Nunez Nepomuceno	Dataset analysis, EDA pipeline, class reduction, model training
Diana Katonova	Methodology write-up, preprocessing, experimental design
Md Nahidul Islam	Results analysis, conclusion, ethical considerations

C Code Repository

All code is available at: https://github.com/Davidn1095/skin-disease-classification

The pipeline consists of numbered scripts:

- `01_eda.py` — Exploratory data analysis and visualisation
- `02_baseline.py` — Traditional ML baselines (RF, LogReg, SVM, KNN)
- `03_transfer_learning.py` — ResNet18 / EfficientNet-B0 fine-tuning
- `04_comparison.py` — Model comparison and summary generation

SLURM batch scripts (`*.sbatch`) are provided for HPC execution on GPU nodes.